

LDA math/logic

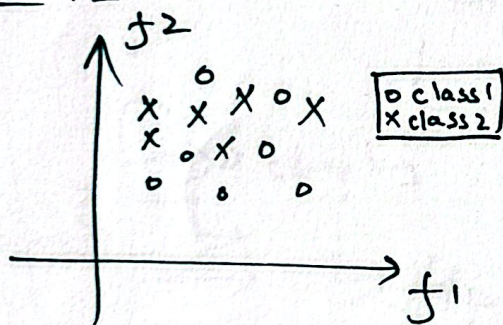
What does LDA do? LDA: Linear discriminant Analysis

Very simply, LDA finds a projection space ($N-1$ dimens.) for N classes such that:

- 1) Distance b/w class means is maximised
- 2) Variance ~~between~~ within classes is minimized.

For a 2D classification problem:

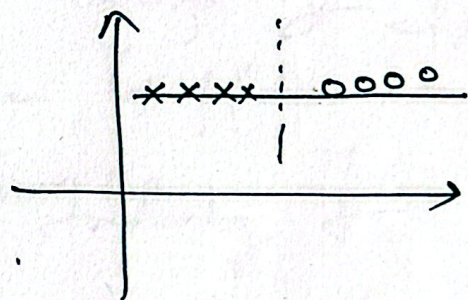
Before



2D plane

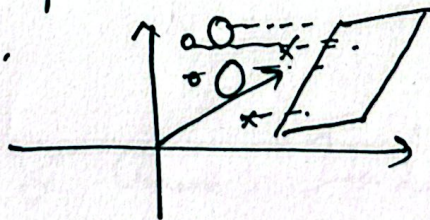
LDA

After: LDA axis



1D LDA axis

For 3 classes, LDA projection space would be a 2D plane and so on...



Take away: LDA is a linear classifier. Finds a linear boundary to separate the classes.

Next step: the math :- D(?)

LDA Math

We approach the math using Fisher criterion:

$$J(w) = \frac{(u_1 - u_0)^2}{\sigma_1^2 + \sigma_0^2} = \frac{A}{B}$$

$\downarrow$
1D formula!

where A is the distance b/w class means and B is the sum of variance of both classes. The idea behind the formula is to have A as big as possible and B as small as possible: maximize mean dist b/w classes
minimize within class variance.

Hold up!! where is the w in the formula regarding $J(w)$?

This is where the idea of finding the LDA projection axis/space comes in! But first, let's extend our 1D formula to work for N dimensions and utilize some Linear Algebra! [$X \Rightarrow$ feature vector!, $u_1, u_0 =$ means, $\sigma_1, \sigma_0 =$ variances of classes 1, 0 & 1.]

$$J(w) = \frac{(u_1 - u_0) \cdot (u_1 - u_0)^T}{(X_1 - u_1) \cdot (X_1 - u_1)^T + (X_0 - u_0) \cdot (X_0 - u_0)^T}$$

$\downarrow$
2 classes

Now, we project this onto the LDA (w) space:

How? $\rightarrow$ we take the dot product of each term in the numerator & the denominator $\Rightarrow$

$$J(w) = \left[w^T \cdot (x_1 - \mu_1) \cdot (w^T \cdot (x_1 - \mu_1))^T \right]$$

$$\left[w^T (x_1 - \mu_1) (w^T (x_1 - \mu_1))^T + w^T (x_0 - \mu_0) (w^T (x_0 - \mu_0))^T \right]$$

using $(A \cdot B^T)^T = B \cdot A^T$

$$J(w) = \left[w^T (\mu_1 - \mu_0) (\mu_1 - \mu_0)^T \cdot w \right]$$

$$\left[w^T (x_1 - \mu_1) \cdot (x_1 - \mu_1)^T w + w^T (x_0 - \mu_0) \cdot (x_0 - \mu_0)^T w \right]$$

$$\Rightarrow \frac{w^T (\mu_1 - \mu_0) \cdot (\mu_1 - \mu_0)^T w}{w^T \left[(x_1 - \mu_1) \cdot (x_1 - \mu_1)^T + (x_0 - \mu_0) \cdot (x_0 - \mu_0)^T \right] w}$$

we will call this S_B

$$w^T \left[(x_1 - \mu_1) \cdot (x_1 - \mu_1)^T + (x_0 - \mu_0) \cdot (x_0 - \mu_0)^T \right] w$$

we will call this S_W

S_B, S_W are now called the scatter matrices.

$S_B \Rightarrow$ between class variance.

$S_W \Rightarrow$ within class variance.

$$J(w) = \frac{w^T S_B w}{w^T S_W w}$$

$\rightarrow$ actually find w , we!

$$\frac{dJ(w)}{dw} = 0$$

generalised eigen value problem!

this gives us $\Rightarrow S_B \cdot w = \lambda S_W \cdot w$

eigen values

$$\Rightarrow S_W^{-1} \cdot S_B \cdot w = \lambda w$$

eigen vectors

... so how are W and λ actually used?

i) we have a feature vector X , $X \in \mathbb{R}^{N \times M}$

where $N \rightarrow$ no. of trials and M is the number of features. ~~For~~ Then,

$\Rightarrow W \in \mathbb{C}^{M \times M} \rightarrow W$ is a $M \times M$ dimensional matrix
 $\downarrow$
possibly complex

$\Rightarrow \lambda \in \mathbb{R}^{M \times 1} \rightarrow M$ values.

$\rightarrow$ the values of λ_i correspond to an axis in w (w_i).

$\hookrightarrow$ these values also tell us how well the classes are separated!!

$\hookrightarrow$ so, we sort the values of λ and then pick the highest values of λ_i 's and get w_i 's corresponding to them.

$\hookrightarrow$ we take $C-1$ w_i 's where C is the number of classes.

$\hookrightarrow W \in \mathbb{C}^{M \times C-1}$

So for 2 classes:

$W \in \mathbb{C}^{M \times 1}$

Small note: The entire derivation is done for 2 classes.

To generalize beyond two classes, SB is computed slightly differently. Instead of using $(u_1 - u_0) \dots$ we use $(u_1 - u_{\text{common}})$ $(u_2 - u_{\text{common}}) \dots$. The S_W variable is extended accordingly.

$$w_{\text{common}} = w_{\text{class 1}} + w_{\text{class 2}} + \dots + w_{\text{class N}}$$

$$w_{\text{common}} = \frac{w_{\text{class 1}} + w_{\text{class 2}} + \dots + w_{\text{class N}}}{N}$$

The code corresponding to this math uses w_{common} .

Consider matrix M of size $N \times M$. S is a vector of size M .

vector M of size M . S is a vector of size M .

(Given) we are given a matrix M of size $N \times M$.
 Here each row of M is a vector of size M .
 We want to find the vector S of size M such that
 the dot product of each row of M with S is the same.
 i.e. for each row i , $M[i, :] \cdot S = c$ for some constant c .
 This is a system of linear equations. We can write it as
 $M \cdot S = c \cdot \mathbf{1}$ where $\mathbf{1}$ is a vector of ones of size N .

vector of size M .

$M \cdot S = c \cdot \mathbf{1}$

for S of size M .

M is $N \times M$.

vector S of size M is unknown. The entire system is given for S of size M .

Since M is a matrix of size $N \times M$, the system is overdetermined if $N > M$.

(usually) we use the least squares method to find the vector S that minimizes the error.